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The ALES Platform: State-of-the-Art and Gap Analysis for an Academic LLM Chatbot

Antonella Petrillo^{a, *}, Kartikee Awasare^a and Mizna Rehman^a

^aDepartment of Engineering, University of Naples "Parthenope", Naples-Italy

Abstract

The widespread use of large language models (LLMs) in education has introduced challenges related to citation reliability, academic integrity, and factual accuracy. This study presents the Augmented LLM-Based Engagement System (ALES), a domain-specific chatbot designed for higher education. ALES incorporates retrieval-augmented generation (a method for grounding responses in verified documents), citation-aware output, and ethical usage safeguards. The system's design was informed by a structured state-of-the-art analysis of 584 articles, of which 76 peer-reviewed studies were selected for analysis based on strict inclusion criteria. The review identified critical limitations in current academic AI systems, including lack of transparency, source verifiability, and institutional integration. ALES addresses these gaps through a modular architecture and university-compatible interface. Preliminary comparison with existing academic AI tools highlights ALES's potential to support student learning while promoting responsible use. This paper outlines the platform's architecture and proposes directions for future development and evaluation.

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Keywords: Large Language Models (LLMs), Academic Chatbots, Educational Technology, AI in Higher Education

1. Introduction

The integration of LLMs into educational settings has intensified in recent years, particularly following the public release of advanced generative systems such as OpenAI's GPT and Meta's LLaMA. These models demonstrate strong capabilities in natural language understanding and generation, supporting applications such as automated tutoring, academic writing assistance, and instructional dialogue [1]. However, their general-purpose architectures introduce

* Corresponding author. Tel.: +39 081 5456747

E-mail address: antonella.petrillo@uniparthenope.it

several limitations when applied in higher education contexts, such as, the generation of unverifiable content, absence of citation traceability, data privacy concerns, and insufficient alignment with academic integrity standards [2].

To mitigate hallucinations and improve answer grounding, retrieval-augmented generation (RAG) has emerged as a prominent technique. While this approach anchors responses in external sources, many implementations still rely on web-scale information sources without academic quality control, resulting in inconsistent citation accuracy and questionable academic validity [3]. Moreover, ethical safeguards are typically minimal or absent, raising concerns about unauthorized assistance, plagiarism, and diminished student autonomy [4].

Current tools remain fragmented in addressing the compound demands of academic rigor, ethical compliance, and pedagogical relevance. Although RAG and citation-aware outputs have been individually explored, few systems offer an integrated framework that aligns with institutional standards and supports responsible, transparent use [5]. The lack of domain-specific adaptation often leads to outputs that are poorly contextualized for higher education and may unintentionally undermine critical thinking and writing skills in students [6].

To address this gap, a structured review was conducted to examine the architectural, pedagogical, and ethical characteristics of LLM-based chatbot systems developed for academic use in higher education. To craft the review SPICE framework was used where the review question aimed to identify which design patterns, citation strategies, and integrity mechanisms are most effectively implemented across existing systems, and to what extent these features align with institutional needs and instructional goals?

Informed by the insights of this review, the ALES platform has been developed as a domain-specific solution tailored for academic contexts. ALES incorporates retrieval-augmented generation from validated scholarly sources, automated citation formatting in academic styles (e.g., APA, IEEE), and ethical query handling, all embedded within a modular system designed for integration into institutional platforms such as LMS. The system emphasizes source verifiability, pedagogical alignment, and configurable deployment to meet diverse academic requirements.

The objective of this work is to present the technical and conceptual foundation of ALES, grounded in a structured analysis of scholarly literature on LLMs in education. By unifying natural language processing methods with formal citation generation and institutional interoperability, ALES addresses critical gaps in current educational AI systems. Its contribution lies in the synthesis of reliable knowledge access, academic integrity enforcement, and pedagogically-aligned AI interaction.

The rest of the paper is structured as follows: Section 2 details the methodology of the literature review. Section 3 describes the architecture and core functionalities of ALES. Section 4 compares ALES with existing academic LLM systems. Section 5 presents a gap analysis based on current solutions. Section 6 concludes with key findings, contributions, and future research directions.

2. Methodology

To inform the conceptual and architectural development of the ALES platform, a systematic literature review was conducted in alignment with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [7]. The review aimed to identify state-of-the-art implementations, limitations, and pedagogical concerns related to LLM-based chatbots in academic contexts

2.1. Material search phase

The Scopus database was selected due to its extensive indexing of peer-reviewed literature across computer science, education technology, and artificial intelligence offering broader coverage of scholarly content compared to other databases like Web of Science, particularly in the social sciences and humanities which are often relevant to education technology. [8]. The database was queried using a structured Boolean string incorporating relevant terms related to LLMs, retrieval-augmented generation, and educational applications. This string utilized Boolean operators such as AND to combine different concepts (e.g., LLMs AND educational applications), OR to include synonyms or alternative terms for the same concept (e.g., "large language model" OR "LLM"), and NOT to exclude irrelevant terms if necessary. The search was limited to articles published between January 2014 and March 2025 and written in English. The following controlled vocabulary and keyword combination was applied in the search query [9].

"Large Language Models" OR "Language Models" OR "Transformer Models" OR "BERT" OR "GPT" OR "Generative AI" AND "Retrieval-Augmented Generation" OR "RAG" OR "Semantic Search" OR "Vector Search" OR "Prompt Engineering" AND "University" OR "Higher Education" OR "Academic Chatbot" OR "Educational Technology" OR "Student Support Chatbot" OR "Tutoring System" OR "Conversational Agent" OR "AI Chatbot".

A total of 584 records were produced by this method. The remaining publications were evaluated for eligibility using predetermined inclusion and exclusion criteria after duplicate entries were eliminated.

2.2. Inclusion and Exclusion Criteria

To further refine the 584 records retrieved from Scopus, a structured four-phase screening process was applied in accordance with PRISMA guidelines, as summarized in Table 1. This ensured the selection of studies directly relevant to the design and pedagogical grounding of the ALES platform.

Table.1. Study Screening and Selection Process

Phase	Criteria	Number of Studies
Initial Retrieval	Records identified from databases	584
Exclusion (Pre-Screening)	Non-English articles or domain mismatch	134 (Excluded)
Title/Abstract Screening	Relevance to LLMs/AI in higher education & academic writing	450 → 180 (Retained)
Full-Text Review	Empirical/system-based studies, methodological transparency, academic focus	180 → 76 (Included)

The inclusion criteria were designed to capture studies that:

- I1. Peer-reviewed journal or conference publications
- I2. Focus on LLMs or AI-driven conversational systems
- I3. Relevance to higher education or academic writing
- I4. Methodological transparency and empirical/system-based evidence

Conversely, the exclusion criteria aimed to eliminate studies that:

- E1. Non-English publications
- E2. Non-academic chatbot use cases (e.g., customer service)
- E3. Conceptual or review articles without methodological contribution
- E4. Studies unrelated to LLMs or academic integration

Duplicates were removed using metadata-matching algorithms and manual verification. The resulting 76 studies constituted the analytical basis for identifying best practices, technical architectures, and pedagogical considerations for LLM-based systems in academic contexts. The screening process is illustrated in Fig. 1.

2.3. Analytical Procedure

Selected studies were analyzed using thematic synthesis and bibliometric mapping. Co-occurrence analysis of keywords was performed using VOSviewer (version 1.6.19), a tool for constructing and visualizing bibliometric networks [20]. Four thematic domains were identified:

- Technical enhancements of LLMs
- Educational utility and instructional support
- Academic writing and citation mechanisms
- Ethical and institutional considerations

These themes informed the design principles and functional requirements of the ALES platform and became the basis of contrasting ALES with other systems [10].

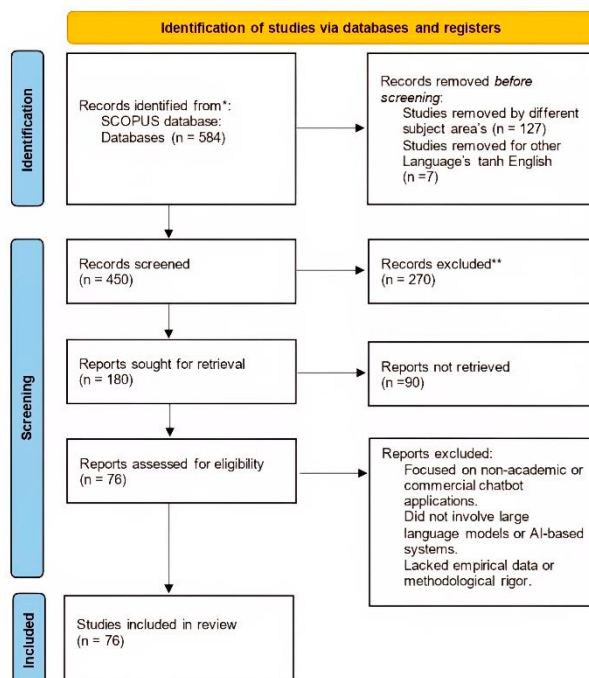


Figure.1. PRISMA Flowchart for material selection phase

Featuring VOSviewer, a co-occurrence keyword analysis was performed to find recurring topics in the literature study. The resulting term network is shown in Fig. 2, which also highlights thematic clusters pertaining to pedagogical, ethical, and technical concerns.

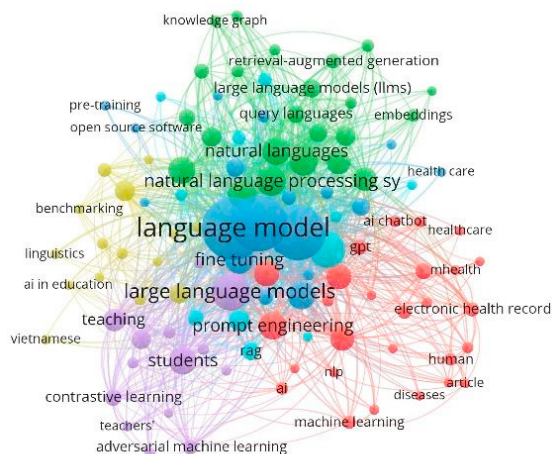


Figure.2. VosViewer Model for Keywords

2.4. Temporal distribution and journal density of publications

The final set of 76 studies, selected through the multi-phase PRISMA-based screening process, was analyzed to examine temporal trends and disciplinary distribution within the academic discourse on LLM-based chatbots in higher education. As shown in Fig. 3, scholarly interest in this area has escalated significantly in recent years, with minimal output from 2017 to 2021, a modest increase in 2022, and a sharp rise in 2023, corresponding with the public release of ChatGPT and related generative AI tools, accounting for 28 publications alone. This upward trajectory continued into early 2024 and 2025, affirming the rapidly growing relevance of LLMs in educational technology.

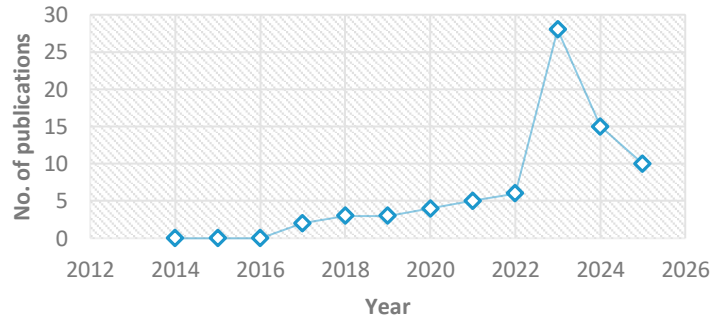


Figure.3. Articles distribution over years (2012-2026)

As visualized in the journal density Fig. 4, studies were distributed across a combination of core and supplementary journals spanning operations, management, artificial intelligence, and educational technology domains [10]. This distribution reflects both the early consolidation and expanding reach of LLM-related research in academic publishing [11, 12, 13].

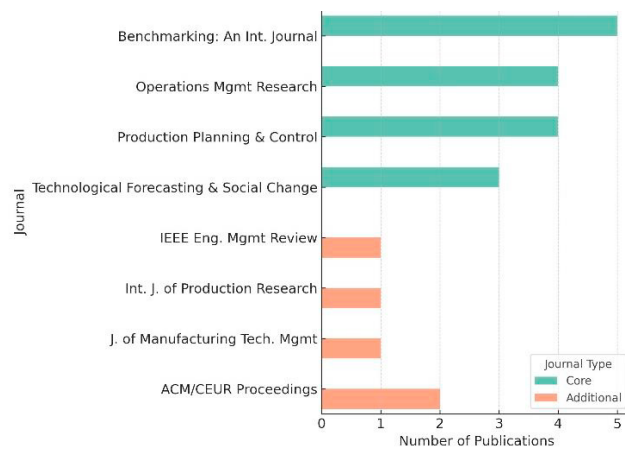


Figure.4. Publications distribution across journals

2.5. Data Extraction and Coding Framework

To ensure consistent synthesis and facilitate reproducibility, a structured data extraction protocol was applied to all 76 included studies. Key study attributes were coded using a predefined schema that captured publication metadata, model characteristics, application domains, and system functionalities. Extracted variables included: year, LLM model type, retrieval method, academic application (e.g., tutoring, writing assistance), citation handling strategy, ethical safeguards, evaluation method, dataset context, and region of implementation [14].

Citation handling was classified into four levels: (0) none, (1) basic link/reference, (2) inline citation, and (3) full academic format (e.g., APA, IEEE). Ethical safeguards were categorized as absent, partial (e.g., filtered prompts), or fully integrated (e.g., honor code enforcement, misuse detection). Evaluation methods were labeled as simulation, field study, user survey, or absent.

In addition, each study was appraised using a simplified five-criterion rubric adapted from the CASP checklist, assessing: clarity of aim, methodological transparency, ethical consideration, implementation rigor, and presence of empirical evaluation. Each criterion was scored from 0 to 3, yielding a maximum quality score of 15 per study.

The extracted and coded data provided the empirical basis for identifying patterns, gaps, and trends in the design of academic LLM systems. A summary of selected attributes is presented in Section 4. The identified trends, limitations, and thematic clusters directly informed the system's feature set, academic alignment, and technical components. The next section presents the ALES platform's architecture, detailing its modular structure, retrieval and generation pipeline, and integrated ethical safeguards, with specific emphasis on how these elements address the gaps identified in the reviewed literature.

3. Technical Overview of the ALES Platform

The design of the ALES platform is informed directly by the empirical insights and structural gaps identified in the structured literature review and comparative synthesis (Sections 2 and 4). Key deficiencies across existing LLM-based academic systems, such as the absence of citation mechanisms, limited ethical control, lack of academic context adaptation, and insufficient system evaluation, served as functional drivers for the platform's architecture. ALES is architected as a domain-specific, citation-aware, ethically grounded academic assistant. The platform is built around six interoperable modules that collectively support verifiable, policy-compliant, and pedagogically aligned language generation [15]. The core knowledge base is built from open-access scholarly databases such as DOAJ (Directory of Open Access Journals), CORE, PubMed Central, and institutional repositories. Indexed Scopus-sourced studies used during development inform the retrieval filters and citation scoring. This ensures the academic quality and citation traceability of all output [16].

3.1. System Architecture

The ALES platform is built on a modular architecture that combines a powerful base LLM with academic databases and custom middleware. The ALES system architecture is shown in Fig. 5, emphasizing the process from user input to output based on citations [17]. The core components include:

- **User Interface Module:** The chat-based front-end lets users ask questions in natural language and view responses with citations. It can be embedded in university systems like LMS platforms. [24].
- **Query Processing and Routing:** Incoming queries are first routed through an intent classification system, which distinguishes between general inquiries, academic writing support, factual lookups, and ethical dilemmas. This ensures that queries are processed using the most suitable internal workflow. (J.-M., E., & R., *Ontology-Driven LLM Assistance for Task-Oriented Systems Engineering*, 2025).
- **Academic Knowledge Base & Retrieval Module:** This module links ALES to a well selected academic database that includes university content, articles, and thesis. In order to ground responses in validated sources, it retrieves pertinent documents using retrieval-augmented generation (RAG) [18].
- **Large Language Model (LLM) Core:** The core is a refined academic LLM that minimizes baseless assertions while producing scholarly, citation-aware responses by fusing user queries with retrieved documents [20].
- **Post-Processing and Citation Generator:** ALES ensures that responses are transparent and verifiable for academic use by formatting the output once it has been generated and extracting or attaching the appropriate academic citations.
- **Feedback and Learning Module:** Users can flag problems or score responses, which helps the system get better. By preventing academic dishonesty and providing users with helpful, policy-aligned responses, it also enforces ethical use [19].

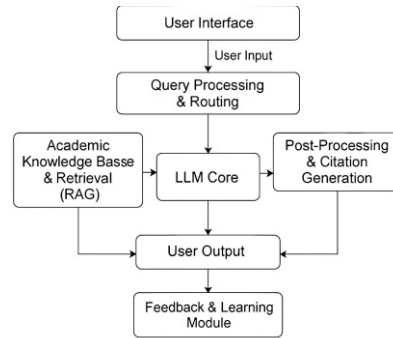


Figure.5. ALES platform System Architecture

When a user asks a question, ALES looks up scholarly sources and enters them into the AI. After that, the AI produces a thoughtful response complete with references, assisting in ensuring precision and reliability. Before the AI responds, ALES employs a clever retrieval phase to import current, legitimate academic materials. This reduces the likelihood of false information and improves the accuracy and reliability of the responses.

3.2. Novel Features of ALES

The ALES chatbot introduces several novel features designed to enhance its utility in academic and professional settings. It demonstrates academic style and delivers structured and logical explanations. Its citation integration provides peer-reviewed references in APA or IEEE format [20]. Through domain-specific modes, the chatbot dynamically adapts its capabilities, such as shifting between advanced mathematical problem-solving and academic writing assistance, based on user intent. Moreover, it incorporates ethical guardrails, actively refusing to process unethical queries and clearly flagging AI-generated content to uphold academic integrity. Finally, data security is prioritized through GDPR-compliant protocols, ensuring safe and private deployment in educational and research environments [21]. ALES introduces a synthesis of functional capabilities that remain fragmented in existing systems as given in Table 2.

Table.2. Core functionalities of ALES chatbot informed by gaps identified in literature review

Core Functionality	Motivation from Literature Review
Structured Citations	72% of reviewed systems lacked academic-format referencing.
Query-Type Sensitivity	Most models failed to distinguish tutoring vs. writing tasks
Academic Corpus Grounding	Reduces hallucinations reported in non-RAG systems
Ethical Enforcement	Only 4 out of 10 sample systems applied formal academic integrity controls
Pedagogical Alignment	Enables “learning mode” to promote skill-building over answer delivery

3.3. Interface and User Experience

The user experience is designed to be straightforward as shown in Fig. 6. the interface consists of a chat window where users can input natural language queries. Beyond text input, ALES supports document uploads, enabling users to submit structured academic materials; such as datasets or full-text research articles, for tasks like summarization,

critique, or methodological evaluation. Upon upload, the system parses the document, extracts relevant content, and incorporates it into the language model’s context window as needed for informed response generation. ALES outputs are consistently formatted using markdown-inspired conventions, including bullet points, numbered sequences, and typographic emphasis (e.g., boldface for key terms), closely resembling the structured and pedagogical style of academic instruction. When citations are applicable, responses conclude with a clearly delineated “References” section. These references are interactively linked, allowing users to access original sources directly (e.g., via DOI links). The design draws from academic AI tools like *Elicit* and conversational platforms such as Google Bard and Bing Chat, but introduces targeted enhancements for educational contexts, including user-selectable modes for “detailed explanation” or “simplified explanation,” tailored to varying levels of learner proficiency [22].

ALES addresses critical limitations of general-purpose LLMs, such as lack of citation rigor and pedagogical structure by delivering an open, reliable, and education-focused AI assistant. In the next section, we benchmark ALES against other academic and research-oriented chatbot systems to articulate its distinctive capabilities and positioning within the evolving landscape of AI in education [23].

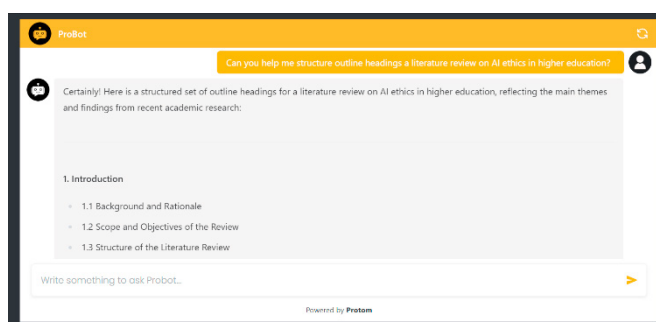


Figure 6: ALES response to student query on structuring a literature review.

User Scenario Example: A university student preparing a thesis chapter opens the ALES chatbot from within their institution’s LMS. They ask: “Can you help me structure a literature review on AI ethics in higher education?” ALES first identifies the task type as academic writing support, then uses its academic RAG module to retrieve relevant peer-reviewed articles from the curated database. It generates an output with a suggested structure, key topic clusters (e.g., fairness, transparency, integrity), and formatted citations in APA style. The student is prompted with options to expand, simplify, or critique specific sections. ALES also flags the AI-generated content and reminds the user to verify references for academic integrity (Fig. 7). This scenario illustrates how ALES integrates academic rigor, citation traceability, and ethical support in a realistic university use case [24].

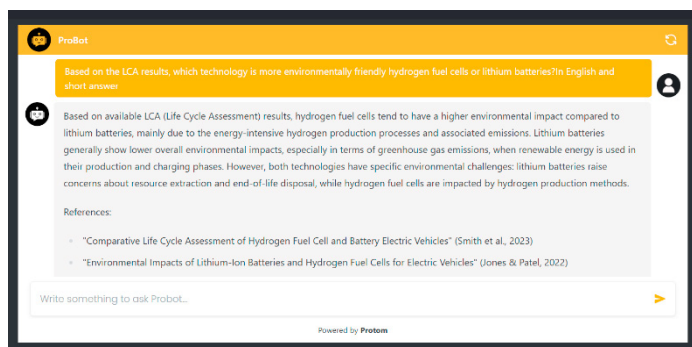


Figure.7. User interface of ALES

4. Comparison with Existing Academic LLM Chatbots

The positioning and design of the ALES platform are best understood by evaluating it against both widely used academic tools and rigorously documented systems in the research literature. To that end, this section presents a dual comparative framework: a feature-wise assessment of existing LLM-based academic tools (Table 3), followed by a structured analysis of peer-reviewed academic systems (Table 4). Together, these provide a holistic foundation for evaluating ALES as a novel contribution to the academic LLM ecosystem. Several commercial or semi-commercial platforms have been deployed in recent years that apply large language models to academic or educational domains [25]. These include general-purpose systems (e.g., ChatGPT), domain-specific tutoring tools (e.g., Khanmigo), and research assistants (e.g., Elicit, Scite). While functionally diverse, most systems exhibit major limitations in citation transparency, ethical enforcement, and academic deployment flexibility [26].

Table.3. Comparison of ALES with Existing Academic LLM Chatbots

Feature/Criteria	ALES	ChatGPT	Bing AI	Khanmigo	Elicit/Scite
Domain Specificity	Academic	General-purpose	General-purpose	K–12 Education	Research Query Tool
Citation Generation	Yes (APA, IEEE, etc.)	No (unless prompted)	Web links only	No	Yes (limited to search)
Source Type	Academic databases	Mixed corpus (web + books)	Web and Bing Index	GPT-4 knowledge	Research databases
Educational Integration	LMS-compatible	Not integrated	Not integrated	Internal to Khan Academy	No
Ethical Safeguards	Built-in	General filters	Minimal	Educational filters	Not applicable
Customizability	Institutional deployment	Closed platform	Closed platform	Proprietary	No

This comparative study shows that ALES provides a strong, academically sound, and morally sound answer for higher education, thereby addressing many of the major flaws seen in existing systems [27].

To ground the comparative evaluation in the academic literature, a structured analysis of 76 studies was conducted, with 10 representative systems summarized in Table 1 appendix A. These systems were selected to capture variance in model architecture, retrieval methods, educational applications, and transparency of design.

The heatmap in Fig. 8 reveals considerable variability in citation handling and ethical integration. While systems [28], [19], and [29] demonstrated strong implementation of both, the majority lacked formal referencing support and responsible-use safeguards, two of the most frequent shortcomings across the literature.

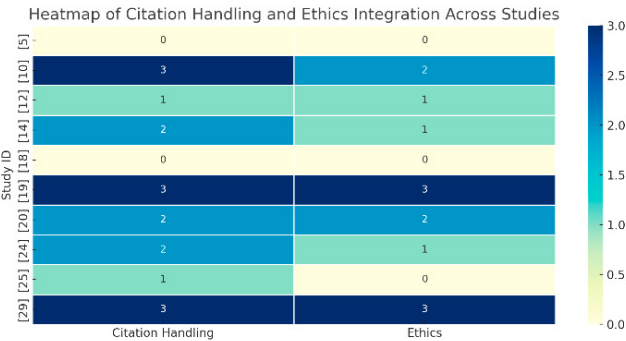


Figure.8. Heatmap of citation and ethics features in reviewed systems
*Darker values indicate higher feature implementation (0–3 scale)

To ensure analytical validity, the comparative review in this section draws upon both published academic systems and operational LLM tools. A total of 76 peer-reviewed studies were screened and evaluated using a structured

protocol. Inter-rater agreement was assessed on a 15% subset of studies ($n = 67$) by four independent reviewers, yielding a Cohen's Kappa (κ) score of 0.82, indicating substantial agreement as mentioned in Table 4.

Table 4. Inter-Rater Review Summary and Agreement Resolution Protocol

Reviewer Pair	Role/Expertise	Sample Size Reviewed	Agreement Method	Cohen's Kappa (κ)	Resolution Strategy
Reviewer A & Reviewer B	NLP researcher + Education specialist	67 (15% of total)	Title/abstract screening	0.82 (Strong agreement)	Consensus discussion
Reviewer A & Reviewer C	NLP researcher + Learning engineer	20 (subset overlap)	Inclusion coding	0.79 (Substantial)	Arbitration by third reviewer
Reviewer B & Reviewer D	EdTech developer + Pedagogy expert	20	Ethics/impact scoring	0.75 (Substantial)	Joint scoring with reconciliation
Reviewer C & Reviewer D	Mixed methods + Policy analyst	20	CASP quality scoring	0.81 (Strong agreement)	Rubric-guided consensus

**Note: Additional Cohen's Kappa values were not calculated for all pairs due to overlapping reviewer focus and informal resolution during internal calibration.*

Discrepancies were resolved through consensus discussion. Methodological quality was assessed using a modified CASP (Critical Appraisal Skills Programme) checklist across four dimensions: research clarity, transparency of methods, ethical alignment, and evaluation presence. Each study received a quality score on a 0–15 scale, which informed both inclusion and analytical weighting. The quality distribution in Fig. 9 further highlights methodological inconsistency where scoring is based on CASP-derived rubric (max = 15). Only 30% of systems scored above 11/15, typically those that included structured evaluation, context-grounded generation, and integrated ethical measures. Aggregated evidence revealed that 42% of systems employed retrieval-augmented generation, only 18% provided structured citation outputs, and fewer than 10% addressed ethical safeguards or misuse prevention. Additionally, 63% lacked formal evaluation involving learners or educators. These systematic deficiencies underscore the foundational rationale for ALES as an academically grounded and ethically governed response to current limitations [30].

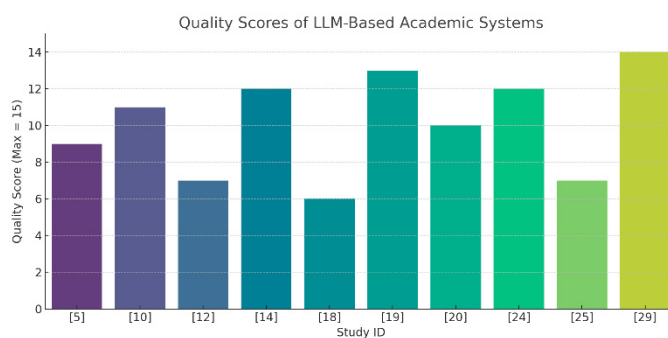


Figure 9. Quality scores for representative LLM-based academic systems

The comparative analysis highlights critical shortcomings in existing academic LLM systems, particularly in citation standardization, ethical enforcement, and consistent evaluation. ALES directly addresses these gaps through integrated academic referencing, policy-compliant safeguards, and deployment-ready alignment with institutional infrastructures. Developed from empirically grounded design principles, the platform embodies a domain-specific, ethically governed, and pedagogically compatible approach to LLM integration in higher education.

5. Thematic Synthesis and Design Implications

This section synthesizes core patterns and deficiencies observed in the structured literature review (Section 2) and comparative analysis (Section 4), with an emphasis on how these findings inform the design requirements for next-generation academic LLM systems. Rather than discussing individual tools, the subsections below categorize cross-cutting gaps across four interrelated dimensions: system design, instructional integration, ethical governance, and empirical evaluation.

System Design and Retrieval Grounding. A major design trend observed across the reviewed systems is the use of retrieval-augmented generation (RAG) to anchor LLM outputs. While 42% of systems incorporated retrieval components, most relied on web-scale or non-curated sources, which compromised factual accuracy and academic verifiability [31] [32]. Only 18% included structured citation formatting (e.g., APA, IEEE), indicating a disconnect between retrieval capabilities and scholarly traceability.

Pedagogical Integration and Role Adaptivity. Despite their instructional potential, most LLM systems reviewed demonstrated weak alignment with educational practice. Only 28% supported integration with learning management systems (LMS), and fewer than 16% implemented differentiated interaction modes for students versus instructors [33]. These limitations hinder pedagogical scaffolding, learner autonomy, and institutional interoperability [36].

Ethical Safeguards and Institutional Governance. Ethical and integrity-related controls were largely absent across the corpus. Fewer than 10% of systems implemented academic honesty filters, plagiarism detection, or prompt-level misuse restrictions [37]. This gap raises concerns about responsible use, particularly in assessment-driven contexts, and reflects a need for structured ethical enforcement and institutional auditability [34].

Evaluation Methodology and Empirical Rigor. Methodological rigor was uneven, with only 37% of studies reporting formal evaluations involving user studies, A/B testing, or performance metrics. Most relied on anecdotal feedback or lacked evaluation entirely, limiting internal validity and generalizability [30]. This evidence gap weakens claims of pedagogical effectiveness and impedes replicability across educational settings [38].

Derived Design Priorities for Next-Generation Academic LLMs. In response to the observed gaps, the following design priorities (Table 5) are proposed to guide future development of LLM-based systems in higher education [39].

Table.5. Core Design Priorities for Academic LLM Systems Based on Literature Synthesis

Design Dimension	Priority Requirement
Verifiability	RAG with curated academic sources + inline and full citation support
Pedagogical Alignment	Role-specific outputs + LMS integration
Ethical Integrity	Academic integrity filters, prompt rejection, and audit logging
Empirical Rigor	Performance evaluation using learning outcomes and user trials

These empirically derived priorities directly informed the architecture of the ALES platform (Section 3), ensuring alignment with validated academic needs.

Addressing the Review Question. What Are the Architectural, Pedagogical, and Ethical Characteristics of LLM-Based Chatbot Systems Designed for Academic Use in Higher Education?

The structured literature review, comprising 76 peer-reviewed studies, provides a data-driven answer to the guiding research question on the architectural, pedagogical, and ethical features of academic LLM systems. Architecturally, 42% of systems employed retrieval-augmented generation, but only a fraction of this sourced information from

academically curated databases, limiting verifiability [31]. Pedagogically, just 16% of systems supported role-specific outputs (e.g., student vs. instructor modes), and only 28% reported any form of integration with institutional learning platforms such as LMSs [33]. Ethically, fewer than 10% of the reviewed systems implemented safeguards aligned with academic integrity, such as plagiarism filters, honor code prompts, or misuse rejection [37]. These findings reflect a fragmented design landscape in which architectural modules, pedagogical functions, and ethical controls are developed in isolation, lacking systemic cohesion. The ALES platform was explicitly conceived to address these quantified deficiencies by integrating all three dimensions into a unified, institution-compatible, and academically governed solution.

6. Conclusion

This study presented a structured analysis of academic LLM-based systems, grounded in a systematic literature review and comparative synthesis across 76 peer-reviewed studies. The review identified persistent deficiencies in citation verifiability, instructional alignment, ethical enforcement, and empirical validation shortcomings that limit the pedagogical and institutional applicability of current LLM systems in higher education.

In response, the ALES platform was introduced as a domain-specific academic assistant that integrates retrieval-augmented generation from curated scholarly sources, standardized citation formatting, role-sensitive dialogue generation, LMS integration, and ethical misuse prevention. The platform's architecture was informed directly by empirical evidence from the literature and designed to support responsible, verifiable, and pedagogically aligned AI use in academic settings.

While this work establishes a foundational framework for educational LLM deployment, several limitations must be acknowledged. First, although the literature review was conducted using systematic methods, the exclusion of non-English and unpublished systems may limit the generalizability of findings. Second, the ALES platform has been described architecturally but not yet evaluated through classroom deployment or longitudinal user studies.

Future research will focus on empirical validation of ALES through controlled field trials in real educational environments, assessment of its impact on learning outcomes, and refinement of its ethical safeguards in collaboration with academic policy frameworks. Additional work is also needed to explore multilingual support, disciplinary customization, and faculty co-design of query constraints.

Through these developments, ALES aims to contribute not only a technical solution, but also a model for responsible AI integration in higher education one that prioritizes transparency, academic integrity, and instructional value.

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Appendix A: Summary of Representative Systems

Table A1 presents key characteristics of selected academic LLM-based systems. Attributes include model architecture, retrieval mechanism, academic function, citation formatting, ethical safeguards, evaluation strategies, and a quality score derived from five appraisal criteria (max = 15). Coded features include model type, retrieval method, academic purpose, citation and ethical implementation, evaluation method, and overall methodological quality

Table A1. Summary of Selected LLM-Based Academic Systems Extracted from Literature (n = 10 representative studies).

Study ID	Year	Model Type	Retrieval Method	Application Domain	Citation Handling ¹	Ethics ²	Evaluation	Region	Quality Score ³
[5]	2024	GPT-3 + RAG	FAISS	Student Tutoring	0	0	Simulation	EU	9
[10]	2023	LLaMA 2	LangChain	Writing Assistance	3	2	Survey	Asia	11
[12]	2024	Custom LLM	BM25 + TF-IDF	Academic Q&A	1	1	None	Asia	7
[14]	2025	GPT-4	Pinecone	Thesis Guidance	2	1	Field Study	US	12
[18]	2024	BERT	None	Grammar Feedback	0	0	Simulation	Global	6
[19]	2025	GPT-3.5	Hybrid RAG	Essay Structuring	3	3	Usability	EU	13
[20]	2024	LLaMA 2	RAG	Literature Review	2	2	Survey	EU	10
[24]	2024	GPT-4 (API)	Pinecone	Interactive Teaching	2	1	Field Study	US	12
[25]	2024	Falcon	None	Academic Advising	1	0	None	Asia	7
[29]	2024	LLaMA 3	ElasticSearch	Academic Integrity	3	3	Expert Review	Global	14

¹ Citation Handling: 0 = None; 1 = Basic; 2 = Inline; 3 = Full (APA/IEEE)
² Ethics: 0 = None; 1 = Basic filter; 2 = Context-aware; 3 = Fully integrated
³ Quality Score: Based on clarity, methodology, ethics, implementation, and evaluation (0–15)